

**FDM 170A: FUNDAMENTALS OF DIGITAL MEDIA PRODUCTION**  
**University of California, Santa Cruz**  
fall term 2024, online

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Book a 15 minute appointment here:  
<https://warrensack.youcanbook.me/>  
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**COURSE CATALOG DESCRIPTION**

Introduction to the conceptual and technical fundamentals of making digital media. Covers principles of digital image manipulation, basic web authoring, and interface design through projects that introduce production techniques and methods.

**APPROACH**

This course takes the approach that digital art can be a means to remake and rethink cinema. We are going to think about how film editing can be practiced by coding, how the web provides an excellent medium for composing multiscreen films, and how we can use programming to make interactive, database films. The outstanding question of digital media is how can we translate the language of art (film and other arts) — issues of composition, interpretation, editing, montage, and narrative — into the language of code and vice versa? We will pay attention to the art and media histories of video and film and exam how ideas and theories of montage and editing of images, especially moving images, can be written as code. Projects will be created for the web with HTML, CSS, and JavaScript. By the end of the course you will be making simple, interactive videos and have the conceptual and technical tools to embark on making sophisticated, interactive videos.

This version of 170A was co-developed by Professor Warren Sack and Abram Stern, a recent PhD graduate and current lecturer for the Film + Digital Media Department.

**REQUIREMENTS**

There are eight required projects for this course. Seven are defined. For the eighth and final project, I ask that you pick one of the seven earlier projects and produce a second version of that project. The seven defined projects are these:

1. *Re-sequencing*: Gather at least twenty of Eadweard Muybridge's pre-cinematic images and re-sequence and compose them into a story on a webpage.
2. *Collage and Concrete Poetry*: Find a concrete poem that you like and reproduce it with CSS and HTML; or find or write a poem that was not conceived of as a concrete poem and make it into one by reformatting it with CSS and HTML.
3. *Montage*: Using some JSON and JavaScript, re-edit some of director Sergei Eisenstein's *Battleship Potemkin* (1925).
4. *Instructions for Drawings*: By hand and with code, JavaScript, execute one of artist Sol LeWitt's wall drawings.
5. *Multi-Screen Interactive*: Using HTML, CSS, and JavaScript create a nine screen interactive composition on a webpage.
6. *{Narrative} Structures*: Write a story grammar in artificial intelligence planning operators expressed as JSON that can generate variants of a set of stories.
7. *Database Cinema*: Assemble and annotate (with JSON) a database of video clips and then repurpose your story grammar so that it is used to automatically generate narrative montages.
8. *Repeat*: Reiterate one of the above seven projects.

For each project there are two graded pieces of work: the completed project and one paragraph of critique that you will write for each of the members in your working group for each of their submitted projects. The class will be divided into working groups of four students each, so you will be responsible for posting a critique in response to the completed projects of three other students eight times over the course of the term.

**What is critique?** Critique is an essential part of all arts education. Normally, a critique would be held in class with everyone in attendance in order to foster an understanding of how your classmates and your instructor understand the artwork you are creating. I distinguish two kinds of critique. Negative critique is a form of complaint or argument against the artwork or the artist. Positive critique is a form of advocacy, an attempt to try to understand what the artist is trying to do and give helpful suggestions about how to make it better. Personal comments about the artist are out of bounds. Positive critiques must focus on what you can see and describe about the artwork. **It does not matter whether you like an artwork or don't like it, so it's best if you don't mention your opinion in your critiques — even if you like an artwork!** This is the hardest part of writing a critique: remember it is not about what you like or don't like it is about what you can see and what you can experience. Focus on describing what you can see, what you can understand about the artwork. **The best critiques are thus, paradoxically, almost purely descriptive.** For this course, we will focus on performing positive critiques. Positive critiques do not begin with an overall negative evaluation of the work. Even the most rigorous positive critique always begins with a sentence expressing understanding for what the artist has achieved or is attempting to achieve with the artwork. It is possible to give a positive critique even if you do not like the artwork you are critiquing! In fact, I advocate that you do not say, in your critiques, the phrase "I like this." In fact, try not to use any of the pronouns I, me, my, or myself. Instead of saying what you like, try to communicate an understanding

for what the artist (*aka* your peer in the class) is trying to do. Any negative comments, must include suggestions for how to make the work better. A very useful phrase is the following: “The artwork could be strengthened in the following ways...” [continued with a set of constructive suggestions for modifying the artwork].

For each assigned project, the instructor has prepared six items:

- a. Begin by screening the project’s video introduction to the art or media history pertinent to the project work, including descriptions and links to important artworks and theories that could provide you with ideas and inspirations for your own work. **These videos will also provide you with vocabulary you can use when you write critiques for your peers.** These introductions are short (10 minutes to 30 minutes) videos. The scripts for the dialogues of the videos is also provided as a PDF so that you can access any URLs mentioned or shown in the videos.
- b. A set of readings for each project will be available on the course website as PDFs. After watching the introductory video, you will need an hour or so to complete the readings. For some assignments, there is also a recommended screening, either a film or websites hosting moving image artworks. Screenings can require up to 2 hours.
- c. For each project, there is an extended description detailing what you will need to do and, when finished, what you will need to post for review by the instructor and by your working group. These descriptions are in the Assignments section of the course website.
- d. A link is available to you for a talk-through of the code (HTML, CSS, and/or JavaScript) that you will use or can modify and extend for your project work. A talk-through is an online format for teaching coding originally developed by John Resig for Khan Academy. Each talk-through is displayed in a web browser as a window on the left, that contains the code; a window on the right, that displays the (usually graphical) output of the code; and an audio track set below the code and output windows. By starting the audio track, one can watch the whole thing as if it is just a standard video: the narrator will fill the lefthand window with code and you can see the results in the righthand window. **Resig’s innovation, however, allows the viewer to interrupt at any moment and change the code in the lefthand window** (and, thereby, change the output displayed in the righthand window). The 170A instructors have developed their own version of the software for creating talk-throughs and authored a talk-through specific to each of the projects. Our talk-throughs typically take 15 or 30 minutes to complete.
- e. A link to Code Pen containing the code you will be editing for the assignment. You will copy this code into your own Code Pen account and edit it there. You will post your final versions by sharing a link to it with your working group.
- f. A critique prompt is also posted for every project in the Discussions section of the course website. These prompts are there to help you think about how to write your one paragraph critiques for each of the members of your working group.

For each project, you will review these six items, create an artwork in response to the project description, and write three critiques, one for the work of each member of your

working group. We estimate that the work for each project will take between 10 and 12 hours.

For those of you who do not already have the knowledge of HTML, CSS, and JavaScript necessary for completion of a project, we recommend you devote additional time to working through a series of Khan Academy talk-throughs to gain that background knowledge. At the end of this document is a detailed schedule that recommends which Khan Academy talk-throughs you should complete before which projects. If you know the material addressed in the Khan Academy talk-throughs, or prefer to learn that material in another way, please feel free to skip the Khan Academy talk-throughs.

## LINKS

You will be posting your finished work to a site called Code Pen (<https://codepen.io/your-work>). To get started, you need to set up your own account on the Code Pen site.

**Each of the projects has two URLs (links):** one for the talk-through, and one for a version of the code that I have posted to Code Pen. To fit into the Code Pen environment some changes had to be made to the code, so the two versions are not identical, the one on Code Pen and the version shown in the talk-through. Although you can play with the code for the project in the talk-throughs' editor windows, you will want to work on the code in Code Pen.

(Or, if you are already fluent with another development environment (e.g., VSCode and Github) you can download the code from the Code Pen instructor's account, work on it in your environment, and then post your final version to Code Pen in your account.)

Here are all the URLs/links for all of the projects:

FILM 170A.1: Re-Sequencing

TalkThrough: <https://people.ucsc.edu/~wsack/FILM170A/TalkThroughs/resequencing.html>

Pen: <https://codepen.io/wsack/pen/GRVKBEK>

FILM 170A.2: Concrete Poetry

TalkThrough: <https://people.ucsc.edu/~wsack/FILM170A/TalkThroughs/concrete.html>

Pen: <https://codepen.io/wsack/pen/XWvrBBY>

FILM 170A.3: Montage

TalkThrough: <https://people.ucsc.edu/~wsack/FILM170A/TalkThroughs/montage.html>

Pen: <https://codepen.io/wsack/pen/qBeWyJv>

FILM 170A.4: Drawings

TalkThrough: <https://people.ucsc.edu/~wsack/FILM170A/TalkThroughs/drawings.html>

Pen: <https://codepen.io/wsack/pen/jOgNovO>

FILM 170A.5: Multiscreen

TalkThrough: <https://people.ucsc.edu/~wsack/FILM170A/TalkThroughs/multiscreen.html>

Pen: <https://codepen.io/wsack/pen/ExqYzGX>

FILM 170A.6: Structures

TalkThrough: <https://people.ucsc.edu/~wsack/FILM170A/TalkThroughs/structures.html>

Pen: <https://codepen.io/wsack/pen/xxvxGOE>

FILM 170A.7: Database Cinema

TalkThrough: <https://people.ucsc.edu/~wsack/FILM170A/TalkThroughs/database.html>

Pen: <https://codepen.io/wsack/pen/WNVNMXm>

## SCHEDULING

A detailed schedule for this course can be found at the end of this document.

This course is designed to be taken online and asynchronously, although there are some hours during the week when you are invited to meet with the instructor during office hours on Zoom. During office hours (on Zoom) you can meet with the instructor one-on-one or in small groups. The times for office hours are listed at the top of this syllabus along with a Zoom link. All office hours will normally be held on Zoom.

You will be assigned a working group, four students per group. You will be writing short (one paragraph) critiques of each project produced by the other three people in your working group. If your group only has three people in it, you will write a critique for the other two students in your group and one for yourself — in the third person, as if you were someone else other than yourself. The working groups are visible in the “Discussions” section of Canvas. If you would like to move to another working group than the one to which you have been assigned, please email the instructor.

If you are new to all of the material in this course (both the media art historical references and coding) and you want to get an A, you will probably need to devote 15-20 hours per week to the course. The Educational Opportunity Programs (EOP) office has a time budgeting tool to help plan your schedule that may be useful: <https://eop.ucsc.edu/resources/advising-timemanagement.html>

**Projects are due each week on Friday.** You must finish your projects before midnight Pacific Time (i.e., California time). I mention the time zone assuming that some students will not be in California and may be taking the course from a different time zone.

**Submission of your projects:** You will submit your projects by posting them to your Code Pen account. You will need to share the link to the project with all of your group members and with the instructor. Post the link to the Canvas-based discussion for the project. So, your first post to the discussion will be several days before you write your

critiques and post them to the discussion. Your first post will simply be a link to the project on your Code Pen account.

You will be penalized for late submissions of your projects primarily because you need to make them available for the members of your working group to critique. **Critiques are due each week on Monday.** You will not be penalized for posting your critique late if the reason is that a member of your working group has submitted their project late. If a project has been posted late, critiques are only tardy 24 hours after the project has been posted. You will not be penalized for not posting a critique of a project that one of your working group members does not hand in. If you don't want to wait for a classmate's project to be posted late, you can always write a critique of your own work as your third critique for the week.

**Submission of your critiques:** You will post your critiques to Canvas-based discussions where the instructor and all of your working group members can read them.

## GRADING

Projects:  $8 \times 9.5 = 76$

Critiques:  $24 \times 1 = 24$

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100

A project submitted late, will have one point (aka 1% of the final course grade) deducted for every day or part of a day it is late.

## Grade Scale

A+ 100% to 99.99%

A < 99.99% to 95%

A- < 95% to 90%

B+ < 90% to 87%

B < 87% to 83%

B- < 83% to 80%

C+ < 80% to 77%

C < 77% to 73%

C- < 73% to 70%

D+ < 70% to 67%

D < 67% to 63%

D- < 63% to 60%

F < 60% to 0%

## **DRC REMOTE ACCOMMODATIONS**

The Disability Resources Center reduces barriers to inclusion and full participation for students with disabilities by providing support to individually determine reasonable academic accommodations. Operations continue via remote appointments. If you have questions or concerns about exam accommodations or any other disability-related matter, email the DRC Schedulers at [drc@ucsc.edu](mailto:drc@ucsc.edu) for an appointment.

## **DISTRIBUTION OF COURSE NOTES**

Please note that selling, preparing, or distributing for any commercial purpose course lecture notes or video or audio recordings of any course is explicitly forbidden by campus policy, unless authorized by the University in advance – for instance, by the Disability Resource Center, which can assign students who need one a note taker with a laptop or other recording device – and explicitly permitted by the course instructor in writing. (DRC note taking accommodations should, and usually will, be announced to the instructor in advance in formal communication from the DRC.)

## **DIFFICULT MATERIAL STATEMENT**

In Film + Digital Media courses you will often be assigned films or other material that could contain difficult ideas, uncomfortable language, or graphic depictions of sex or violence. You will be asked to treat these portrayals critically, to consider what is being expressed by the maker, or to examine the potential social impact, and to evaluate the works in a given context. Instructors are happy to speak with you and might direct you to CAPS (Counseling and Psychological Services) at 831-459-2628 or the Disability Resource Center ([drc@ucsc.edu](mailto:drc@ucsc.edu) or 831-459-2089) should you need additional support in order to do your best work.

## **UCSC PRINCIPLES OF COMMUNITY**

In this class, we abide by the UCSC Principles of Community, so please familiarize yourself with these important principles: <https://www.ucsc.edu/about/principles-community.html>

## **ACADEMIC DISHONESTY**

Academic integrity is the cornerstone of a university education. Academic dishonesty diminishes the university as an institution and all members of the university community. It tarnishes the value of a UCSC degree. All members of the UCSC community have an explicit responsibility to foster an environment of trust, honesty, fairness, respect, and responsibility. All members of the university community are expected to present as their original work only that which is truly their own. All members of the community are expected to report observed instances of cheating, plagiarism, and other forms of

academic dishonesty in order to ensure that the integrity of scholarship is valued and preserved at UCSC.

In the event a student is found in violation of the UCSC Academic Integrity policy, he or she may face both academic sanctions imposed by the instructor of record and disciplinary sanctions imposed either by the provost of his or her college or the Academic Tribunal convened to hear the case. Violations of the Academic Integrity policy can result in dismissal from the university and a permanent notation on a student's transcript.

For the full policy and disciplinary procedures on academic dishonesty, students and instructors should refer to the Academic Integrity page (<https://ue.ucsc.edu/academic-misconduct.html>) at the Division of Undergraduate Education.

## **INCLUSIVITY STATEMENT**

As an instructor at UC Santa Cruz, I value equality of opportunity, human dignity, and racial/ethnic/cultural diversity. Be assured that I will promote a safe and conducive environment for learning. In accordance with University policy, I will not tolerate discrimination or harassment on the basis of race, color, religion, national origin, ancestry, sex, age, marital status, familial status, sexual or gender orientation, disability, or status as a disabled veteran or a veteran of the Vietnam era. In addition to the University's policy, and within the bounds of the course, I do not discriminate on the basis of political or religious beliefs, and I also make every effort to avoid discrimination on the basis of class or income. This means that you do not have to agree with me or the assigned readings in order to do well in this course. You are, however, obligated to demonstrate an understanding of the course material. If there is something I can do to make the class more hospitable, please let me know.

## **TITLE IX**

The university cherishes the free and open exchange of ideas and enlargement of knowledge. To maintain this freedom and openness requires objectivity, mutual trust, and confidence; it requires the absence of coercion, intimidation, or exploitation. The principal responsibility for maintaining these conditions must rest upon those members of the university community who exercise most authority and leadership: faculty, managers, and supervisors.

The university has therefore instituted a number of measures designed to protect its community from sex discrimination, sexual harassment, sexual violence, and other related prohibited conduct. Information about the Title IX Office (<https://titleix.ucsc.edu/>) the online reporting link (<https://uctitleix.i-sight.com/portal/Santa%20Cruz>), applicable campus resources, reporting responsibilities, the UC Policy on Sexual Violence and Sexual Harassment (<https://policy.ucop.edu/doc/4000385/SVSH>), and the UC Santa Cruz Procedures for Reporting and Responding to Reports of Sexual Violence and Sexual Harassment can be found at (<https://titleix.ucsc.edu/>).

The Title IX Office is actively responding to reports and requests for consultation. If you are not currently working with someone in the office and want to make a report/request a consult, you can expect the fastest response by using the online reporting link <https://uctitleix.i-sight.com/portal/Santa%20Cruz> or phone number 831-459-2462

For more information please visit the Title IX Operations under Covid-19 page: <https://titleix.ucsc.edu/about/titleix-covid19.html>

## LAND ACKNOWLEDGEMENT

The land on which USC is built is the unceded territory of the Awaswas-speaking Uypi Tribe. The Amah Mutsun Tribal Band, comprised of the descendants of indigenous people taken to missions Santa Cruz and San Juan Bautista during Spanish colonization of the Central Coast, is today working hard to restore traditional stewardship practices on these lands and heal from historical trauma.

## SCHEDULE

Every week you are invited to attend office hours with the instructor alone or with your working group. You might also consider having your working group convene for an hour or so every week without the instructor. Most weeks you will have a project due on **Friday** (before midnight) and critiques of your classmates' project work will be due on **Mondays** (with one exception due to the holiday on Monday, November 11th).

During the first days of the term, you will need to do a number of things including setting up (free) accounts for both Drop Box (<https://www.dropbox.com/>) and Code Pen (<https://codepen.io/>). A to-do list is given in the first module, "Orientation," posted at the top of the course's Canvas site.

Pamela Fox, former Chief Content Creator for Computer Science at Khan Academy then a lecturer at Berkeley now at Microsoft (<https://www.linkedin.com/in/pamela-fox-5668b1b4/>) has created about 40 hours of talk-throughs to introduce the basics of HTML, CSS, and JavaScript. You may not know any of these basics, in which case it is our recommendation that you work through all of these talk-throughs — although none of the Khan Academy materials are required for this course, and none of your work on Khan Academy will be a part of your grade for this course. These talk-throughs at Khan Academy are simply a very good resource for quickly learning HTML, CSS, and JavaScript.

Keeping this in mind, the following detailed schedule labels the eight projects with due dates and places them within the sequence of Khan Academy talk-throughs with the assumption that the material covered in those talk-throughs (and a few other referenced sites of tutorials) will be familiar to you before you start the course projects. Each Khan Academy talk-through takes about five minutes to complete.

1. Khan Academy: Intro to HTML: <https://www.khanacademy.org/computing/computer-programming/html-css/intro-to-html/v/making-webpages-intro>
  - 1.1.Welcome to the web!
  - 1.2.HTML basics
  - 1.3.Quick tip: HTML tags
  - 1.4.Challenge: Write a Poem
  - 1.5.HTML: Text emphasis
  - 1.6.Challenge: You can learn text tags
  - 1.7.HTML: Lists
  - 1.8.Challenge: Your learning list
  - 1.9.HTML: Images
    - 1.9.1.Challenge: A picture-perfect trip
2. Khan Academy: Intro to CSS: <https://www.khanacademy.org/computing/computer-programming/html-css/intro-to-css/pt/css-basics>
  - 2.1.CSS Basics
  - 2.2.Quick tip: Selecting by tag name
  - 2.3.Challenge: Colorful creature
  - 2.4.CSS: Selecting by id
  - 2.5.Challenge: Seasonal ids
  - 2.6.CSS: Selecting by class
  - 2.7.Challenge: Apples and bananas classes
  - 2.8.Using simple CSS selectors
  - 2.9.Project: Travel webpage
3. Khan Academy: More HTML tags: <https://www.khanacademy.org/computing/computer-programming/html-css/html-tags-continued/pt/html-links>
  - 3.1.HTML links
  - 3.2.Challenge: Links you love
  - 3.3.HTML internal links
  - 3.4.Challenge: Jump around
  - 3.5.HTML tables
  - 3.6.Challenge: The dinner table
  - 3.7.HTML comments
  - 3.8.Project: Recipe book
4. **Project 1: Re-sequencing** due before midnight, Friday, October 4
5. **Critiques for Project 1** due before midnight, Monday, October 7
6. Khan Academy: CSS text properties: <https://www.khanacademy.org/computing/computer-programming/html-css/css-text-properties/v/css-zen-garden>
  - 6.1.CSS Zen Garden
  - 6.2.CSS font-family property
  - 6.3.Challenge: Fancy font families
  - 6.4.CSS font-size property
  - 6.5.Challenge: Great big font sizes
  - 6.6.CSS font styles and shorthand
  - 6.7.Challenge: Famous font formats
  - 6.8.More CSS text properties

- 6.9.Using CSS text properties
- 6.10.CSS inheritance
- 6.11.Project: Blog
- 7. Khan Academy: Web development tools: <https://www.khanacademy.org/computing/computer-programming/html-css/web-development-tools/a/using-the-browser-developer-tools>
  - 7.1.Using the browser developer tools
  - 7.2.Developing webpages outside of Khan Academy
  - 7.3.Hosting your website on a server
  - 7.4.Hosting your website on Github
- 8. Khan Academy: CSS layout: <https://www.khanacademy.org/computing/computer-programming/html-css/css-layout-properties/pt/css-grouping-elements>
  - 8.1.CSS grouping elements
  - 8.2.Challenge: Group the groupers
  - 8.3.CSS width, height, and overflow
  - 8.4.Challenge: The overflowing ocean
  - 8.5.CSS box model
  - 8.6.Challenge: The boxer model
  - 8.7.CSS position
  - 8.8.Challenge: Position planet
  - 8.9.CSS in the wild: Google Maps
  - 8.10.CSS floating elements
  - 8.11.Challenge: Floating clouds
  - 8.12.Using CSS layout properties
  - 8.13.Planning your webpage
  - 8.14.Project: Event invite
- 9. W3Schools: CSS transform Property: [https://www.w3schools.com/cssref/css3\\_pr\\_transform.asp](https://www.w3schools.com/cssref/css3_pr_transform.asp)
- 10. **Project 2: Collage and Concrete Poetry** due before midnight, Friday, October 11
- 11. **Critiques for Project 2** due before midnight, Monday, October 14
- 12. Khan Academy: More CSS selectors: <https://www.khanacademy.org/computing/computer-programming/html-css/more-css-selectors/pt/using-multiple-css-classes>
  - 12.1.Using multiple CSS classes
  - 12.2.Challenge: A classy gallery
  - 12.3.Combining CSS class and element selectors
  - 12.4.Challenge: Classes of elements
  - 12.5.CSS descendant selectors
  - 12.6.Challenge: Descendants of Khan
  - 12.7.Grouping CSS selectors
  - 12.8.CSS dynamic pseudo-classes
  - 12.9.Challenge: Grouped animals
  - 12.10.CSS specificity
  - 12.11.CSS specificity rules
- 13. Khan Academy: Other ways to embed CSS: <https://www.khanacademy.org/computing/computer-programming/html-css/more-ways-to-embed-css/pt/using-inline-css-styles>

- 13.1.Using inline CSS styles
- 13.2.Using external stylesheets
- 14.CSS3
  - 14.1.CSS3 selectors: <https://www.teaching-materials.org/deprecated/css3-selectors/>
  - 14.2. CSS3 properties: <https://www.teaching-materials.org/css3-fx/>
- 15.Khan Academy: Further Learning: <https://www.khanacademy.org/computing/computer-programming/html-css/html-css-further-learning/a/webpage-design>
  - 1. Webpage design
  - 2. HTML validation
  - 3. Validating HTML
  - 4. What to learn next
- 16.The <video> tag
  - 1. W3Schools: [https://www.w3schools.com/tags/tag\\_video.asp](https://www.w3schools.com/tags/tag_video.asp)
  - 2. W3: <https://www.w3.org/2010/05/video/mediaevents.html>
- 17.**Project 3: Montage** due before midnight, Friday, October 18
- 18.**Critiques for Project 3** due before midnight, Monday, October 21
- 19.Khan Academy: Intro to programming: <https://www.khanacademy.org/computing/computer-programming/programming/intro-to-programming/v/programming-intro>
  - 19.1.What is Programming?
  - 19.2.Learning programming on Khan Academy
- 20.Khan Academy: Drawing basics: <https://www.khanacademy.org/computing/computer-programming/programming>
  - 20.1.Making drawings with code
  - 20.2.Quick tip: number scrubbing
  - 20.3.Challenge: Simple snowman
  - 20.4.Drawing more shapes with code
  - 20.5.Challenge: Waving snowman
- 21.Khan Academy: Coloring: <https://www.khanacademy.org/computing/computer-programming/programming/coloring/pt/coloring-with-code>
  - 21.1.Coloring with code
  - 21.2.Quick tip: color picking
  - 21.3.Challenge: Sunny snowy day
  - 21.4.The Power of the Docs
  - 21.5.Project: What's for dinner
- 22.Khan Academy: Variables: <https://www.khanacademy.org/computing/computer-programming/programming/variables/pt/intro-to-variables>
  - 22.1.Intro to Variables
  - 22.2.Using variables
  - 22.3.Challenge: Bucktooth Bunny
  - 22.4.More on Variables
  - 22.5.Challenge: Funky Frog
  - 22.6.Review: Variables
- 23.Khan Academy: Animation basics: <https://www.khanacademy.org/computing/computer-programming/programming/animation-basics/a/what-are-animations>
  - 23.1.What are animations?

- 23.2. Making animations
- 23.3. Challenge: Exploding Sun
- 23.4. Incrementing shortcuts
- 23.5. A shorter shortcut
- 23.6. Challenge: Parting Clouds
- 23.7. Project: Shooting star
- 24. HTML <canvas> tag tutorial: [https://www.w3schools.com/graphics/canvas\\_intro.asp](https://www.w3schools.com/graphics/canvas_intro.asp)
- 25. Khan Academy: Interactive programs: <https://www.khanacademy.org/computing/computer-programming/programming/interactive-programs/pt/mouse-interaction>
  - 25.1. Mouse Interaction
  - 25.2. Challenge: Tasty Tomato
  - 25.3. Challenge: Mouse movement mania
- 26. Khan Academy: Becoming a community coder: <https://www.khanacademy.org/computing/computer-programming/programming/becoming-a-community-coder/a/evaluate-peer-projects>
  - 26.1. Evaluate peer projects
  - 26.2. Ask for help
- 27. Khan Academy: Bonus: Resizing with variables: <https://www.khanacademy.org/computing/computer-programming/programming/resizing-with-variables/a/using-math-expressions-in-js>
  - 27.1. Using math expressions in JS
  - 27.2. Resizing shapes with variable expressions
  - 27.3. Using variable expressions
  - 27.4. Challenge: Brown bear eyes
  - 27.5. Project: Animal attack
- 28. Khan Academy: Text and strings: <https://www.khanacademy.org/computing/computer-programming/programming/text-basics/pt/terrific-text-part-one>
  - 28.1. Terrific Text: Part One
  - 28.2. Challenge: My Favorite Foods
  - 28.3. Terrific Text: Part Two
  - 28.4. Challenge: Mouse Tracker
  - 28.5. Review: text and strings
  - 28.6. Project: Ad design
- 29. Khan Academy: Functions: <https://www.khanacademy.org/computing/computer-programming/programming/functions/pt/functions>
  - 29.1. Challenge: Say Your Name
  - 29.2. Function Parameters
  - 29.3. Challenge: Moles in Holes
  - 29.4. Function Return Values
  - 29.5. Challenge: Calculator
  - 29.6. Local and Global Variables
  - 29.7. Special ProcessingJS functions
  - 29.8. Review: Functions
  - 29.9. Project: Fish tank

- 30. Khan Academy: Logic and if Statements: <https://www.khanacademy.org/computing/computer-programming/programming/logic-if-statements/pt/if-statements>
  - 30.1.If Statements
  - 30.2.Challenge: Bouncy Ball
  - 30.3.More Mouse Interaction
  - 30.4.Challenge: Your First Painting App
  - 30.5.Booleans
  - 30.6.Challenge: Number Analyzer
  - 30.7.Logical Operators
  - 30.8.Challenge: Your First Button
  - 30.9.Challenge: Smarter Button
  - 30.10.If/Else - Part 1
  - 30.11.Challenge: Flashy Flash Card
  - 30.12.If/Else - Part 2
  - 30.13.Review: Logic and if Statements
  - 30.14.Random numbers
  - 30.15.Project: Magic 8-Ball
- 31. Khan Academy: Debugging programs: <https://www.khanacademy.org/computing/computer-programming/programming/debugging-programs/pt/debugging-with-printlns>
  - 31.1.Debugging with println()s
  - 31.2.More debugging tips
- 32. Khan Academy: Looping: <https://www.khanacademy.org/computing/computer-programming/programming/looping/pt/intro-to-while-loops>
  - 32.1.Intro to While Loops
  - 32.2.Using while loops
  - 32.3.Challenge: A Loopy Ruler
  - 32.4.More While Loops: Balloon Hopper
  - 32.5.Challenge: A Loopy Landscape
  - 32.6.For Loops! A New Kind of Loop
  - 32.7.Challenge: Lined Paper
  - 32.8.Nested For Loops
  - 32.9.Review: Looping
  - 32.10.Project: Build-a-House
- 33. Khan Academy: Writing clean code: <https://www.khanacademy.org/computing/computer-programming/programming/writing-clean-code/pt/readable-code>
  - 33.1.Readable Code
  - 33.2.Clarifying with Comments
  - 33.3.Writing clean code
- 34. Khan Academy: Arrays: <https://www.khanacademy.org/computing/computer-programming/programming/arrays/pt/intro-to-arrays>
  - 34.1.Intro to Arrays
  - 34.2.Challenge: Favorite Fruits
  - 34.3.Looping through Arrays
  - 34.4.Challenge: Favorite Animals

- 34.5.Modifying Arrays
- 34.6.Challenge: Constellation Maker
- 34.7.Review: Arrays
- 34.8.Project: Make it rain
- 35. Khan Academy: Objects: <https://www.khanacademy.org/computing/computer-programming/programming/objects/pt/intro-to-objects>
  - 35.1.Intro to Objects
  - 35.2.Challenge: Recipe Card
  - 35.3.Modifying Objects
  - 35.4.Challenge: Picture Painter
  - 35.5.Arrays of Objects
  - 35.6.Challenge: Movie Reviews
  - 35.7.Review: Objects
  - 35.8.Project: Bookshelf
- 36. Khan Academy: Object-Oriented Design: <https://www.khanacademy.org/computing/computer-programming/programming/object-oriented/pt/object-types>
  - 36.1.Object Types
  - 36.2.Challenge: Double Rainbow
  - 36.3.Object Methods
  - 36.4.Challenge: SmileyFace
  - 36.5.Object Inheritance
  - 36.6.Challenge: Flower Grower
  - 36.7.Review: Object-Oriented Design
- 37. Khan Academy: Becoming a better programmer: <https://www.khanacademy.org/computing/computer-programming/programming/good-practices/a/planning-a-programming-project>
  - 37.1.Planning a programming project
  - 37.2.Planning with pseudo-code
  - 37.3.What to learn next
  - 37.4.Help others learn!
- 38. **Project 4: Instructions for Drawings** due before midnight, Friday, October 25
- 39. **Critiques for Project 4** due before midnight, Monday, October 28
- 40. Khan Academy: Get ready to make your webpages interactive: <https://www.khanacademy.org/computing/computer-programming/html-css-js>
  - 40.1.Welcome to making webpages interactive
  - 40.2.Review HTML/CSS
  - 40.3.HTML/CSS recap
  - 40.4.Review JavaScript
  - 40.5.JavaScript recap
- 41. Khan Academy: JS and the DOM: <https://www.khanacademy.org/computing/computer-programming/html-css-js/js-and-the-dom/pt/putting-js-in-a-webpage>
  - 41.1.Putting JS in a webpage
  - 41.2.Challenge: A scripted script tag
  - 41.3.The DOM (Document Object Model)
  - 41.4.Understanding the DOM
  - 41.5.Debugging webpages with the browser console

- 42. Khan Academy: DOM access methods: <https://www.khanacademy.org/computing/computer-programming/html-css-js/html-js-dom-access/pt/finding-elements-by-id>
  - 42.1. Finding elements by ID
  - 42.2. Challenge: ID changer
  - 42.3. Finding multiple DOM elements by tag or class name
  - 42.4. Challenge: Custom name tags
  - 42.5. Finding elements by CSS selector
  - 42.6. Challenge: Query modernizer
  - 42.7. Summary: DOM access methods
- 43. Khan Academy: DOM modification: <https://www.khanacademy.org/computing/computer-programming/html-css-js/html-js-dom-modification/pt/changing-attributes>
  - 43.1. Changing attributes
  - 43.2. Challenge: Avatar attributes
  - 43.3. Changing styles
  - 43.4. Challenge: Style guide
  - 43.5. Changing CSS classes
  - 43.6. Challenge: Classification
  - 43.7. Setting innerHTML and textContent
  - 43.8. Challenge: The inside story
  - 43.9. Creating elements from scratch
  - 43.10. Challenge: Create a solar system
  - 43.11. Summary: DOM modification techniques
- 44. Khan Academy: DOM events: <https://www.khanacademy.org/computing/computer-programming/html-css-js/html-js-dom-events/v/making-webpages-interactive-with-events>
  - 44.1. Making webpages interactive with events
  - 44.2. Adding an event listener
  - 44.3. Challenge: Cat Clicker
  - 44.4. DOM event types
  - 44.5. Using the event properties
  - 44.6. Challenge: Cat-stache
  - 44.7. Processing forms with events
  - 44.8. Challenge: Mad Libs
  - 44.9. Preventing default behavior of events
  - 44.10. Summary: DOM events
- 45. **Project 5: Multi-Screen Interactive** due before midnight on Friday, November 1
- 46. **Critiques for Project 5** due before midnight, Monday, November 4
- 47. Khan Academy: Using JS libraries in your webpage: <https://www.khanacademy.org/computing/computer-programming/html-css-js/using-js-libraries-in-your-webpage/a/whats-a-js-library>
  - 47.1. What's a JS library?
  - 47.2. Using a JS library: Slideshow library
  - 47.3. Where are JS libraries hosted?
  - 47.4. Using JS libraries
  - 47.5. The world of JS libraries

- 47.6. Which JS library should you use?
48. Khan Academy: Further learning: <https://www.khanacademy.org/computing/computer-programming/html-css-js/html-css-js-further-learning/a/html-css-js-further-learning-what-to-learn-next>
- 48.1. What to learn next
49. **Project 6: {Narrative} Structures** due before midnight on Friday, November 8
50. **Critiques for Project 6** due before midnight, Tuesday, November 12 (This is a Tuesday deadline because Monday, the 11th is Veteran's Day.)
51. **Project 7: Database Cinema** due before midnight on Friday, November 15
52. **Critiques for Project 7** due before midnight, Monday, November 18
53. **Project 8: Repeat** second draft of one of the previous assignments: due before midnight on Friday, November 22
54. **Critiques for Project 8** due before midnight, Monday, November 25